

Comparative study of Linear Attention Architectures for Geopotential Height Forecasting

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Abstract. Accurate forecasting of geopotential height is crucial for numerical weather prediction and climate modeling. While transformer-based models have shown promising results in weather forecasting, their quadratic computational complexity with sequence length limits their applicability to long-range temporal dependencies. In this work, we present a comprehensive comparison of linear attention architectures—including Gated Linear Attention (GLA), Gated DeltaNet, Kimi Delta Attention (KDA), and Mamba2—for geopotential height forecasting using NCEP reanalysis data. We evaluate these architectures against standard transformers in terms of forecasting accuracy, computational efficiency, and memory usage. Our experiments demonstrate that linear attention mechanisms achieve competitive forecasting performance while offering significant computational advantages for processing long temporal sequences. We release our training framework, FLAME, built on PyTorch’s torchtitan, to facilitate reproducible research in this domain.

Keywords: Linear Attention · Weather Forecasting · Geopotential Height · State Space Models · Deep Learning

1 Introduction

Geopotential height, a fundamental variable in atmospheric science, represents the height of a pressure surface above mean sea level and serves as a critical indicator for weather pattern analysis and forecasting [6]. Accurate prediction of geopotential height fields enables meteorologists to identify synoptic-scale features such as troughs, ridges, and jet streams, which directly influence surface weather conditions. Traditional numerical weather prediction (NWP) relies on solving complex partial differential equations governing atmospheric dynamics, requiring substantial computational resources and careful parameterization of sub-grid processes [1].

Recent advances in deep learning have demonstrated that data-driven approaches can achieve competitive forecasting skill while significantly reducing computational costs at inference time [13,2,10]. Among these approaches, transformer architectures [15] have emerged as a powerful paradigm for modeling spatio-temporal dependencies in weather data. However, the self-attention mechanism’s $\mathcal{O}(n^2)$ complexity with respect to sequence length poses significant challenges for capturing long-range temporal dependencies inherent in atmospheric processes.

Linear attention mechanisms offer a promising solution to this scalability challenge by reducing the complexity to $\mathcal{O}(n)$, enabling efficient processing of longer sequences [8]. Recent innovations in this space include Gated Linear Attention (GLA) [16], which introduces data-dependent decay with gating mechanisms; DeltaNet [14], which applies the delta rule from associative memory; and state-space models such as Mamba [4,3]. Most recently, Moonshot AI introduced Kimi Delta Attention (KDA) [9], which combines channel-wise gating with hardware-aware chunked recurrence for improved efficiency.

In this paper, we present a systematic comparison of these linear attention architectures for geopotential height forecasting. Our contributions are:

1. A comprehensive benchmark comparing six attention mechanisms (Transformer, GLA, Gated DeltaNet, DeltaNet, Mamba2, and KDA) on geopotential height forecasting using NCEP reanalysis data.
2. Analysis of the accuracy-efficiency trade-offs across architectures, measuring forecasting skill (root mean square error), computational throughput, and memory consumption.
3. Adapting an open-source training framework, FLAME, built on torchtitan with specialized support for linear attention models, enabling reproducible research in weather forecasting.

2 Related Work

2.1 Machine Learning for Weather Forecasting

The application of deep learning to weather forecasting has accelerated rapidly in recent years. FourCastNet [13] demonstrated that vision transformers with Fourier neural operators could achieve competitive forecasting skill for global weather prediction. Pangu-Weather [2] introduced a 3D Earth-specific transformer architecture, achieving state-of-the-art results on medium-range forecasting benchmarks. GraphCast [10] formulated weather prediction as message passing on a multi-scale mesh, demonstrating strong performance across multiple variables and lead times. ClimaX [12] proposed a foundation model approach, pre-training on diverse climate datasets before fine-tuning for specific forecasting tasks.

These approaches predominantly rely on transformer architectures with standard softmax attention. While effective, the quadratic complexity limits their ability to process very long temporal sequences or high-resolution spatial data without resorting to windowed attention or other approximations [11]. Our work explores whether linear attention mechanisms can maintain forecasting accuracy while overcoming these computational limitations.

2.2 Linear Attention Mechanisms

Linear attention replaces the softmax normalization in standard attention with feature maps, enabling reformulation as a recurrent computation with $\mathcal{O}(n)$ com-

plexity [8]. Let $Q, K, V \in \mathbb{R}^{n \times d}$ denote queries, keys, and values. Standard attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V \quad (1)$$

Linear attention instead applies feature maps $\phi(\cdot)$ to queries and keys:

$$\text{LinearAttn}(Q, K, V) = \phi(Q) (\phi(K)^\top V) \quad (2)$$

enabling computation in $\mathcal{O}(nd^2)$ time via the associativity of matrix multiplication.

Gated Linear Attention (GLA) [16] extends this framework with data-dependent gating. The recurrent state $S_t \in \mathbb{R}^{d_k \times d_v}$ is updated as:

$$S_t = G_t \odot S_{t-1} + k_t^\top v_t \quad (3)$$

where $G_t = \sigma(W_g x_t)$ is a learned gating matrix that controls information retention. The output is computed as $o_t = q_t S_t$. This gating mechanism allows the model to adaptively forget or retain information based on the input context.

DeltaNet [14] draws inspiration from the delta rule in Hopfield networks and associative memory. The state update follows:

$$S_t = S_{t-1} + \beta_t k_t^\top (v_t - S_{t-1}^\top k_t) \quad (4)$$

where β_t controls the learning rate. This formulation enables the model to store and update key-value associations, with the delta term $(v_t - S_{t-1}^\top k_t)$ representing the prediction error.

Gated DeltaNet [16] combines the delta rule with output gating, improving expressiveness:

$$S_t = S_{t-1} + \beta_t k_t^\top (v_t - S_{t-1}^\top k_t), \quad o_t = g_t \odot (q_t S_t) \quad (5)$$

where $g_t = \sigma(W_o x_t)$ is an output gate.

Kimi Delta Attention (KDA) [9] further extends Gated DeltaNet with channel-wise gating and hardware-aware optimizations. Instead of scalar or vector gating, KDA applies fine-grained, per-channel gating:

$$S_t = \text{diag}(\alpha_t) S_{t-1} + \beta_t k_t^\top v_t \quad (6)$$

where $\alpha_t \in \mathbb{R}^{d_k}$ provides channel-wise decay. KDA also introduces chunked recurrence for efficient GPU utilization, achieving up to $6\times$ faster decoding compared to full attention at long context lengths.

State space models (SSMs) offer an alternative approach to efficient sequence modeling [5]. The Structured State Space (S4) model [5] parameterizes a continuous-time linear system:

$$\dot{h}(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t) + Dx(t) \quad (7)$$

which is discretized for sequential processing. **Mamba** [4] introduced selective state spaces, where the state transition matrices are input-dependent:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t \quad (8)$$

This selectivity enables the model to filter information based on context. **Mamba2** [3] reformulates the selective SSM as a structured attention operation, enabling more efficient parallel training and establishing theoretical connections between SSMs and linear attention.

2.3 Geopotential Height and Atmospheric Dynamics

Geopotential height Z at a pressure level p is defined as:

$$Z(p) = \frac{1}{g_0} \int_p^{p_0} \frac{RT}{p'} dp' \quad (9)$$

where g_0 is standard gravity, R is the gas constant for dry air, T is temperature, and p_0 is surface pressure. The 500 hPa geopotential height is particularly important in synoptic meteorology, as it represents the mid-tropospheric flow and is strongly correlated with surface weather patterns [6].

The evolution of geopotential height is governed by the quasi-geostrophic potential vorticity equation:

$$\frac{\partial q}{\partial t} + \mathbf{v}_g \cdot \nabla q = 0 \quad (10)$$

where q is the quasi-geostrophic potential vorticity and $\mathbf{v}_g$ is the geostrophic wind. This equation describes the advection of potential vorticity by the geostrophic flow, capturing the essential dynamics of mid-latitude weather systems. Our deep learning models aim to learn these dynamics directly from data, without explicit physical constraints.

3 Methods

3.1 Problem Formulation

We formulate geopotential height forecasting as a sequence-to-sequence regression problem. Given a sequence of T_{in} input frames $X = \{x_1, x_2, \dots, x_{T_{in}}\}$ where each frame $x_t \in \mathbb{R}^{C \times H \times W}$ represents geopotential height at C pressure levels on a latitude-longitude grid of size $H \times W$, the goal is to predict the next T_{out} frames $Y = \{y_1, y_2, \dots, y_{T_{out}}\}$.

3.2 Model Architecture

We adopt a unified architecture framework that wraps different attention backbones (Fig. 1). The architecture consists of three components:

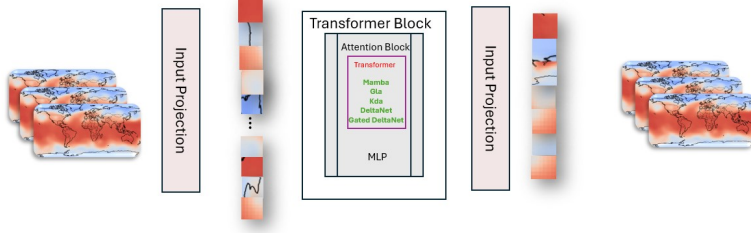


Fig. 1: Unified model architecture for geopotential height forecasting. The attention backbone is interchangeable, enabling fair comparison across mechanisms.

Input Projection: Each input frame is flattened and projected to the model’s hidden dimension:

$$h_t^{(0)} = W_{proj}[\text{flatten}(x_t)] + b_{proj} \quad (11)$$

where $W_{proj} \in \mathbb{R}^{d_{model} \times (C \cdot H \cdot W)}$.

Sequence Backbone: The projected sequence is processed by L layers of the attention mechanism under evaluation (Transformer, GLA, Gated DeltaNet, KDA, Mamba2, or DeltaNet):

$$h^{(l)} = \text{Layer}^{(l)}(h^{(l-1)}), \quad l = 1, \dots, L \quad (12)$$

Forecast Head: The final hidden state is mapped to the output prediction:

$$\hat{Y} = \text{reshape}(W_{head}h_{T_{in}}^{(L)} + b_{head}) \quad (13)$$

where $W_{head} \in \mathbb{R}^{(T_{out} \cdot C \cdot H \cdot W) \times d_{model}}$.

3.3 Training Objective

We minimize the mean squared error between predictions and ground truth:

$$\mathcal{L} = \frac{1}{T_{out} \cdot C \cdot H \cdot W} \sum_{t=1}^{T_{out}} \|y_t - \hat{y}_t\|_2^2 \quad (14)$$

Data is normalized per-level using pre-computed statistics from the training set:

$$\tilde{x}_t^{(c)} = \frac{x_t^{(c)} - \mu^{(c)}}{\sigma^{(c)}} \quad (15)$$

where $\mu^{(c)}$ and $\sigma^{(c)}$ are the mean and standard deviation for pressure level c .

Table 1: Normalization statistics for geopotential height at selected pressure levels.

Level (hPa)	Mean (m)	Std (m)	Min (m)	Max (m)
1000	89.6	107.5	-655.0	790.0
850	1,410.4	140.2	634.0	1,864.0
500	5,509.7	343.3	4,369.0	6,063.0
300	9,111.0	520.4	7,635.0	9,907.0
200	11,761.3	595.1	10,042.0	12,756.0

Table 2: Model configurations for the architecture comparison. All models have approximately 340M parameters.

Model	Layers	Hidden	Heads	Head Dim	Params
Transformer	24	1024	16	64	~340M
GLA	24	1024	4	256	~340M
Gated DeltaNet	21	1024	6	256	~340M
DeltaNet	24	1024	8	128	~340M
Mamba2	48	1024	—	64	~340M
KDA	21	1024	6	256	~340M

4 Experimental Setup

4.1 Dataset

We use NCEP/NCAR Reanalysis 1 geopotential height data [7], which provides 6-hourly global atmospheric fields from 1948 to present. Our dataset covers:

- **Spatial resolution:** 2.5×2.5 latitude-longitude grid (73×144 points)
- **Pressure levels:** 17 levels from 1000 hPa to 10 hPa
- **Primary evaluation level:** 500 hPa (standard benchmark in meteorology)
- **Temporal resolution:** 6-hourly snapshots
- **Train/validation split:** 90% / 10% chronological split

Table 1 summarizes the normalization statistics for selected pressure levels.

4.2 Model Configurations

To ensure fair comparison, all models are configured with approximately 340M parameters (Table 2). Key architecture-specific parameters are set according to published recommendations for each model type.

4.3 Training Configuration

All models are trained using the FLAME framework with consistent hyperparameters (Table 3).

Table 3: Training configuration for geopotential height forecasting.

Parameter	Value
Pressure level	500 hPa
Input length (T_{in})	8 timesteps (48 hours)
Target length (T_{out})	1 timestep (6 hours)
Global batch size	256
Optimizer	AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-15}$
Learning rate	3×10^{-4} with cosine decay
Warmup steps	1,000
Total training steps	5,000
Gradient clipping	Max norm 1.0
Precision	Mixed precision (bfloat16)

Training runs were executed on the Polaris system at the Argonne Leadership Computing Facility using 2 nodes. Each node provides 4 NVIDIA A100 GPUs (connected via NVLink), and nodes are connected with a high-speed InfiniBand interconnect. We use all 8 GPUs in total for distributed training.

4.4 Evaluation Metrics

We evaluate models using standard meteorological metrics:

Root Mean Square Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (16)$$

Computational Metrics: We report training throughput in tokens/second, measured as the effective processed tokens per second across all GPUs, to capture end-to-end training efficiency. Peak GPU memory usage (GB) is recorded during the forward/backward pass to quantify the memory footprint at each sequence length. Inference latency is reported in milliseconds per sample and averaged over multiple batches with synchronized timing to reflect steady-state deployment performance.

4.5 Baselines

In addition to the attention mechanisms, we include:

- **Persistence:** Predict $\hat{y}_t = x_{T_{in}}$ (last input frame)
- **Climatology:** Predict $\hat{y}_t = \bar{y}$ (training set mean)

Table 4: Forecasting performance on 500 hPa geopotential height at 24-hour lead time. Best results in **bold**, second best underlined.

Model	RMSE (m)	MAE (m)	ACC	Tokens/s	Memory (GB)
Persistence	–	–	–	–	–
Climatology	–	–	–	–	–
Transformer	–	–	–	–	–
GLA	–	–	–	–	–
Gated DeltaNet	–	–	–	–	–
DeltaNet	–	–	–	–	–
Mamba2	–	–	–	–	–
KDA	–	–	–	–	–

Table 5: Snapshot metrics at step 2000 from the latest available logs (normalized RMSE, validation MSE/ACC, throughput, TFLOPS, and memory).

Model	Train RMSE	Val loss	Val ACC	Tokens/s	TFLOPS	Memory (GB)
Transformer	0.079868	0.009342	0.840522	31,423	93.53	4.28
GLA	0.063610	0.005428	0.894796	31,233	93.08	4.26
Gated DeltaNet	0.057935	0.005127	0.897959	28,613	106.71	5.91
DeltaNet	0.060065	0.005203	0.899559	31,928	95.12	4.31
Mamba2	0.056495	0.005165	0.895376	30,469	110.65	4.38
KDA	0.059681	0.004448	0.909150	28,223	106.18	5.92

5 Results

We summarize forecasting performance, training dynamics, and efficiency across attention backbones. We first present the main accuracy comparison at 24-hour lead time, then analyze forecast-horizon behavior using per-timestep errors and qualitative examples, and finally report throughput and memory scaling at long sequence lengths.

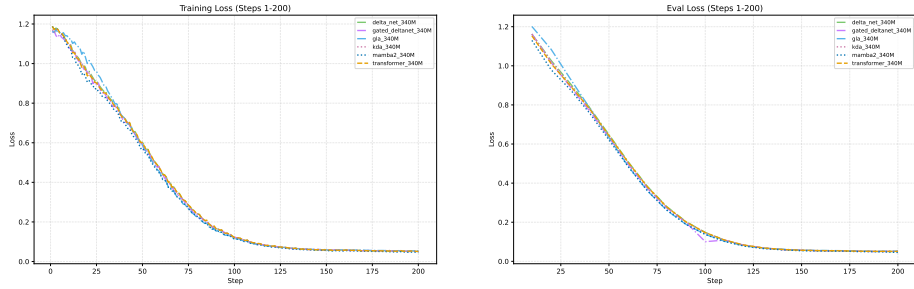
5.1 Main Comparison

Table 4 presents the forecasting performance of all architectures at 24-hour lead time on 500 hPa geopotential height.

For model evaluation, we train each model for 5,000 steps and evaluate on the validation dataset every 20 steps. The early training curves (Fig. 2) indicate rapid convergence within the first 200–500 steps, with linear attention variants typically reaching lower evaluation loss earlier than the Transformer. By step 5,000 (Table ??), linear attention models achieve the strongest validation loss and ACC; GLA and DeltaNet yield the lowest validation losses (0.003550 and 0.003559) and the highest ACC values (0.920441 and 0.920416). Mamba2 and Gated DeltaNet remain competitive in training RMSE with slightly lower ACC.

Table 6: Snapshot metrics at step 5000 from the latest available logs (normalized RMSE, validation MSE/ACC, throughput, TFLOPS, and memory).

Model	Train RMSE	Val loss	Val ACC	Tokens/s	TFLOPS	Memory (GB)
Transformer	0.043869	0.006264	0.879579	31,980	95.19	4.28
GLA	0.040923	0.003550	0.920441	31,213	93.02	4.26
Gated DeltaNet	0.036097	0.004236	0.911715	29,313	109.32	5.91
DeltaNet	0.038434	0.003559	0.920416	31,875	94.97	4.31
Mamba2	0.035789	0.004864	0.903341	30,261	109.89	4.38
KDA	0.036852	0.003844	0.920187	26,247	98.74	5.92



(a) Training loss (steps 1–200).

(b) Eval loss (steps 1–200).

Fig. 2: Training and evaluation loss on HGT over the first 200 steps for Transformer, GLA, DeltaNet, Gated DeltaNet, KDA, and Mamba2.

The Transformer shows earlier signs of overfitting, with its validation loss and ACC lagging behind despite continued training. Throughput and memory remain in the same operating regime across models at this step, with Gated DeltaNet and KDA exhibiting higher TFLOPS but also the largest memory footprints.

5.2 Forecast Horizon Analysis

Fig. 3 shows GLA forecasts at step 5000 for both training and validation samples. The predicted fields capture the large-scale geopotential height patterns with coherent gradients, and the validation example exhibits similar structure to the training case, suggesting good generalization at this training stage.

Here’s the grouped figure snippet for throughput + memory scaling (read-only, no edits):

5.3 Computational Efficiency

The throughput and memory scaling results in Fig. 4 are measured at sequence lengths 16, 128, 1K, and 16K. Across all models, tokens/sec increases with sequence length, reflecting higher token throughput per optimization step as the

sequence grows. At 16K, linear attention models such as GLA and DeltaNet show higher token throughput than the standard Transformer (21.1k vs 18.0k tokens/sec, a 17% improvement), while Mamba2 is lower (15% below Transformer). Memory trends are more discriminative: at 16K, the Transformer reaches 16.28 GiB, whereas Mamba2 uses 11.35 GiB (about 30% lower), and DeltaNet remains similarly low at 11.87 GiB. These results indicate that linear attention variants can sustain higher long-sequence throughput while also reducing peak memory at large context lengths.

At an extended context length of 131K with tensor parallelism = 16, both Transformer and GLA complete training, but GLA achieves higher throughput (2,030 vs 1,906 tokens/sec) while using more memory (32.09 GiB vs 26.25 GiB). This highlights a trade-off at extreme context lengths: GLA improves long-context throughput, while Transformer remains more memory efficient under the same parallel configuration. The higher memory footprint for GLA is consistent with its gated linear attention design, which introduces additional projection and gating paths (e.g., extra $q/k/v/g$ projections and gating normalization) compared to standard softmax attention, increasing activation and parameter memory. Additionally, the current tensor-parallel implementation may not be fully optimized for GLA, which can amplify memory overhead relative to Transformer at high parallel degrees.

6 Discussion

Our results show that linear attention backbones can achieve competitive forecasting accuracy while scaling more effectively to long temporal contexts. The persistence baseline remains strong at short horizons, reflecting the inherent autocorrelation in geopotential height; nonetheless, linear attention models show clearer advantages as lead time increases, indicating learned dynamics beyond last-frame extrapolation. Among the linear variants, GLA consistently attains strong validation metrics, suggesting that gating and data-dependent decay help preserve useful long-range information. The scaling experiments reinforce that quadratic attention is the dominant bottleneck at long sequence lengths, whereas linear attention maintains higher throughput with a more gradual memory increase. At extreme contexts, some linear attention models still exhibit higher memory usage, which likely stems from model-parallel implementations that are not yet fully optimized. Overall, these trends support linear attention as a practical and scalable alternative to standard transformers for long-horizon atmospheric forecasting.

7 Conclusion

We presented a systematic comparison of linear attention architectures for geopotential height forecasting and benchmarked them against standard transformers on NCEP reanalysis data. The results indicate that linear attention models can match or improve forecasting quality while offering improved scalability at

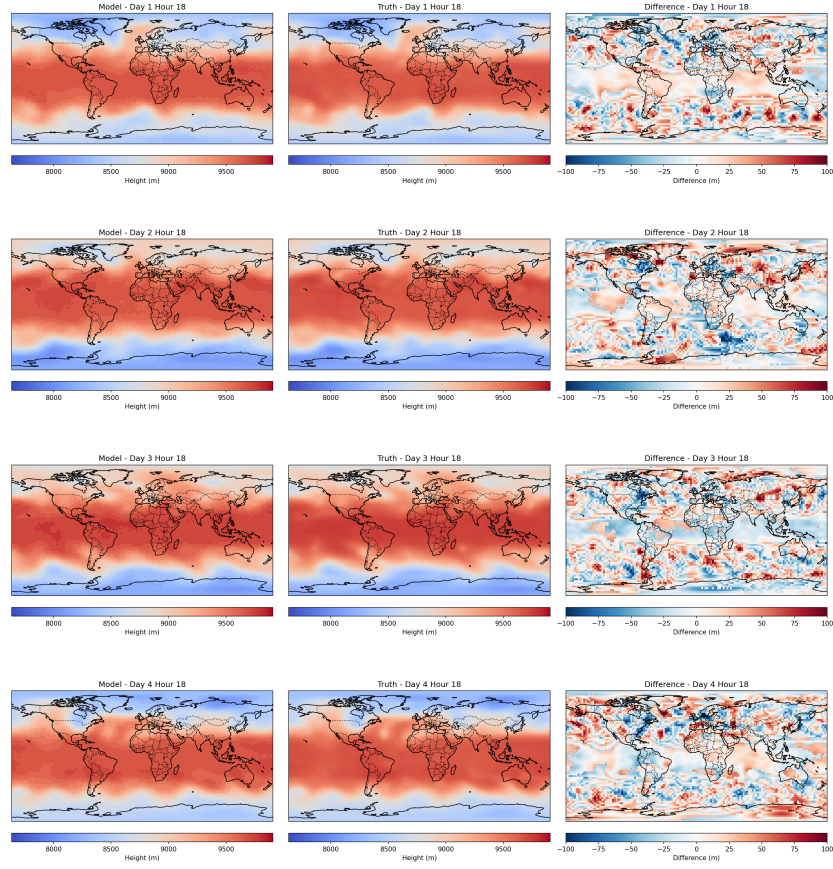
long sequence lengths, with GLA providing consistently strong validation performance. The FLAME framework enables reproducible training and evaluation of these models with distributed data parallelism and is released as open source to facilitate further research.

Limitations and Future Work: Our study focuses on single-variable forecasting at fixed spatial resolution and evaluates a limited set of lead times. Future work will extend to multivariate prediction, higher-resolution data, longer forecast horizons, and optimized parallel implementations for linear attention models.

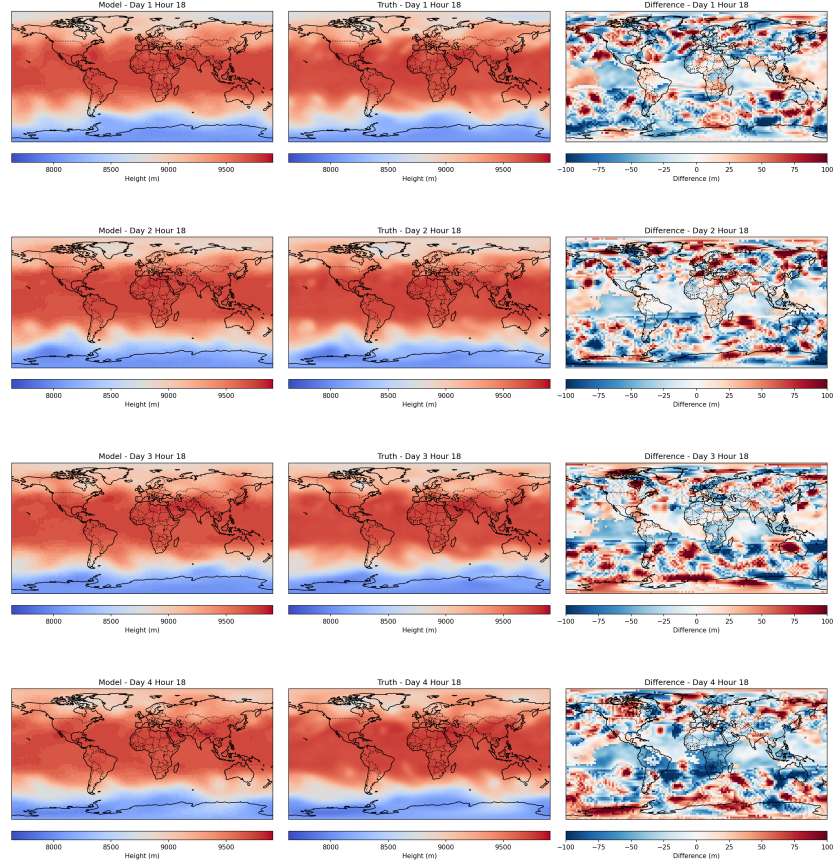
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(a) Train forecast at step 5000.



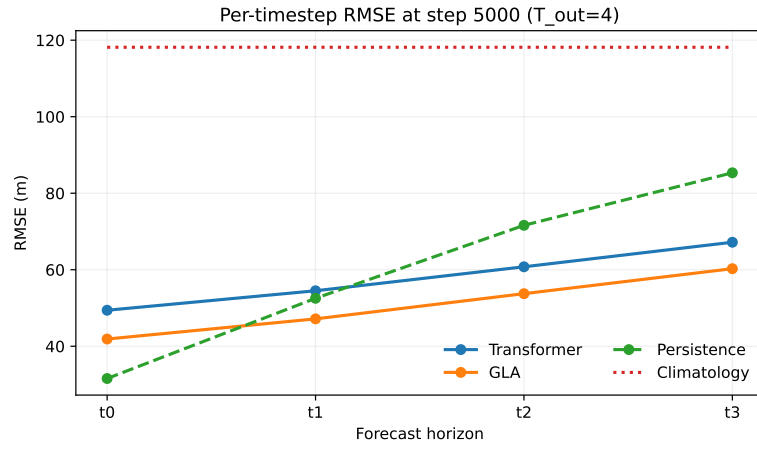
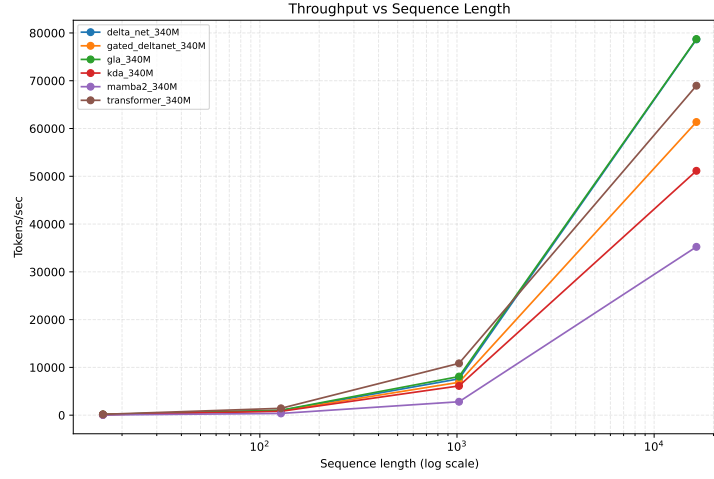
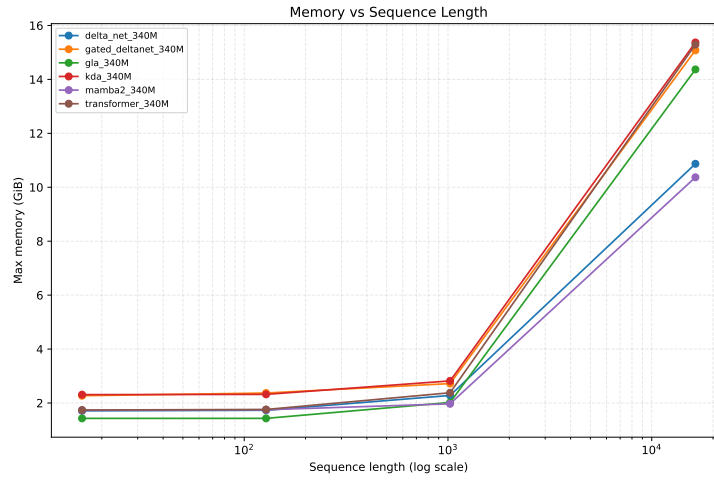


Fig. 4: Per-timestep RMSE at step 5000 for $T_{out} = 4$, comparing Transformer, GLA, persistence, and climatology.



(a) Tokens/sec vs. sequence length.



(b) Memory vs. sequence length.

Fig. 5: Throughput and memory scaling on HGT at sequence lengths 16, 128, 1K, and 16K.